

SplitStack Test Case Specification and Traceability Matrix

Version: 1.0 Date: April 28, 2026 Testing Type: Black-box testing Project: SplitStack shared budgeting and expense management app

1. Purpose

This document defines black-box test cases for the main SplitStack use cases. The tests focus on externally visible behavior, not internal code structure. Each test case includes steps, inputs, and expected outputs. A traceability matrix maps each test case to the use case it verifies.

2. Main Use Cases

Use Case ID	Use Case Name	Actor	Requirement
UC-01	Create Account and Authenticate	Group Member	A user can register, log in, and access the SplitStack workspace.
UC-02	Create and Manage Group	Group Owner/Admin	A user can create a group, set a voting threshold, and invite members.
UC-03	Join Existing Group	Group Member	An invited user can accept or decline a group invitation.
UC-04	Log Expense with Split Options	Group Member	A user can add an expense and split it equally, by percentage, or by custom amounts.
UC-05	Vote on Proposed Expense	Group Member	Expenses above the group threshold require member approval before affecting balances.
UC-06	View Dashboard and Balances	Group Member	A user can view balances, totals, and recent group activity.
UC-07	Settle Outstanding Balance	Group Member	A user can record a settlement payment and reduce outstanding balances.
UC-08	Create and Contribute to Challenge	Group Member	A user can create a savings challenge and add contributions.
UC-09	View Analytics and Insights	Group Member	A user can view spending analytics based on saved expense data.
UC-10	Update Settings and Notifications	Group Member	A user can update settings and mark notifications as read.
UC-11	Chat with AI Assistant	Group Member	A user can ask the SplitStack assistant questions about their data.

3. Test Case Specification

TC-01: Register a New User Account

Field	Details
Related Use Case	UC-01
Objective	Verify that a new user can create a SplitStack account.
Preconditions	The app is running and the email is not already registered.
Inputs	Name: Alex Tester ; Email: alex.test@example.com ; Password: Password123!
Steps	1. Open the SplitStack app. 2. Click Create account . 3. Enter name, email, and password. 4. Submit the registration form.
Expected Output	The account is created successfully. The user is taken to the authenticated workspace and their name/email are displayed.

TC-02: Reject Login with Invalid Credentials

Field	Details
Related Use Case	UC-01
Objective	Verify that the system rejects invalid login attempts.
Preconditions	The app is running.
Inputs	Email: wrong@example.com ; Password: incorrect-password
Steps	1. Open the login page. 2. Enter invalid email and password. 3. Submit the login form.
Expected Output	The user remains on the login screen and an error message is displayed. The workspace is not shown.

TC-03: Create a New Group with Invite

Field	Details
Related Use Case	UC-02
Objective	Verify that a user can create a group and invite another member.
Preconditions	User is logged in.
Inputs	Group Name: Apartment 4B ; Type: Roommates ; Voting Threshold: \$100 ; Invite Email: sam.member@example.com
Steps	1. Go to the Groups page. 2. Open the create group form. 3. Enter group name, type, threshold, and invite email. 4. Submit the form.
Expected Output	The group appears in the user's group list. The voting threshold is saved. The invited email appears as a pending invite.

TC-04: Accept a Pending Group Invite

Field	Details
Related Use Case	UC-03
Objective	Verify that an invited user can join an existing group.
Preconditions	User has a pending group invite.
Inputs	Invite Decision: Accept
Steps	1. Log in as the invited user. 2. Open the pending invite area. 3. Select Accept . 4. Navigate to Groups .
Expected Output	The invite is removed from the pending list. The group appears in the user's group list.

TC-05: Add an Equal Split Expense

Field	Details
Related Use Case	UC-04, UC-06
Objective	Verify that a group member can add an expense split equally among members.
Preconditions	User belongs to a group with at least two members.
Inputs	Group: Apartment 4B ; Description: Groceries ; Amount: \$60 ; Category: Groceries ; Split Method: Equal
Steps	1. Go to Add Expense . 2. Select the group. 3. Enter expense description, amount, and category. 4. Select equal split. 5. Submit the expense. 6. Open the dashboard or group detail page.
Expected Output	The expense is saved and displayed. The amount is divided equally among group members. Dashboard balances update to include the expense.

TC-06: Reject Invalid Percent Split Expense

Field	Details
Related Use Case	UC-04
Objective	Verify that percent split expenses must total 100%.
Preconditions	User belongs to a group with multiple members.
Inputs	Amount: \$100 ; Split Method: Percent ; Member Percentages Total: 90%
Steps	1. Go to Add Expense . 2. Select percent split. 3. Enter percentages that total 90%. 4. Submit the expense.
Expected Output	The system rejects the expense or displays a validation error. The expense is not saved.

TC-07: Create Vote for Expense Above Threshold

Field	Details
Related Use Case	UC-05
Objective	Verify that an expense above the voting threshold creates a pending vote.
Preconditions	User belongs to a group with voting threshold \$100 .
Inputs	Description: New couch ; Amount: \$150 ; Category: Furniture ; Reason: Shared living room furniture
Steps	1. Go to Add Expense . 2. Enter an expense above the group threshold. 3. Submit the expense. 4. Open Group Voting .
Expected Output	A pending vote is created for the expense. The expense is not added to balances until approved.

TC-08: Approve a Pending Expense Vote

Field	Details
Related Use Case	UC-05, UC-06
Objective	Verify that an approved vote causes the expense to count toward balances.
Preconditions	A pending vote exists for an above-threshold expense.
Inputs	Vote Decision: Approve from all required group members
Steps	1. Log in as each voting member. 2. Open Group Voting . 3. Approve the pending expense. 4. Open the dashboard or expense list.
Expected Output	The vote status changes to approved. The expense appears in the expense list. Dashboard balances include the approved expense.

TC-09: Decline a Pending Expense Vote

Field	Details
Related Use Case	UC-05
Objective	Verify that a declined vote prevents an expense from affecting balances.
Preconditions	A pending vote exists for an above-threshold expense.
Inputs	Vote Decision: Decline
Steps	1. Log in as a group member. 2. Open Group Voting . 3. Decline the pending expense. 4. Open the dashboard or expense list.
Expected Output	The vote status changes to declined. The expense does not appear as an approved expense and balances do not include it.

TC-10: View Dashboard Balances

Field	Details
Related Use Case	UC-06
Objective	Verify that the dashboard displays correct financial information.
Preconditions	User belongs to at least one group with saved approved expenses.
Inputs	Existing group and expense data
Steps	1. Log in. 2. Open the Home dashboard. 3. Review total owed, total owed to user, group totals, and recent activity.
Expected Output	Dashboard displays balances and activity based on the user's accepted groups and approved expenses.

TC-11: Record a Settlement Payment

Field	Details
Related Use Case	UC-07
Objective	Verify that a user can record a payment toward an outstanding balance.
Preconditions	User owes another group member money.
Inputs	Payee: owed member; Amount: \$25 ; Method: Venmo ; Note: Partial payment
Steps	1. Open the dashboard. 2. Select the settlement option for an outstanding balance. 3. Enter payee, amount, method, and note. 4. Submit the settlement.
Expected Output	Settlement is saved. Settlement history updates. The outstanding balance is reduced by \$25 .

TC-12: Create a Savings Challenge

Field	Details
Related Use Case	UC-08
Objective	Verify that a user can create a group savings challenge.
Preconditions	User belongs to at least one group.
Inputs	Challenge Name: Save for deposit ; Goal: \$500 ; End Date: future date
Steps	1. Go to Challenges . 2. Enter challenge name, group, goal, and end date. 3. Submit the challenge form.
Expected Output	The challenge is created and displayed with a progress indicator.

TC-13: Add Contribution to Savings Challenge

Field	Details
Related Use Case	UC-08
Objective	Verify that challenge progress updates after a contribution.
Preconditions	A savings challenge exists.
Inputs	Contribution Amount: \$50
Steps	1. Open Challenges . 2. Select an existing challenge. 3. Enter a contribution amount. 4. Submit the contribution.
Expected Output	The contribution is saved. Challenge progress increases by \$50 .

TC-14: View Analytics

Field	Details
Related Use Case	UC-09
Objective	Verify that analytics display spending summaries.
Preconditions	User has at least one approved expense.
Inputs	Existing approved expenses in one or more categories
Steps	1. Log in. 2. Go to Analytics . 3. Review totals, categories, and group summaries.
Expected Output	Analytics page displays spending totals based on saved approved expenses.

TC-15: Update Settings and Mark Notification Read

Field	Details
Related Use Case	UC-10
Objective	Verify that user settings and notification read status can be updated.
Preconditions	User is logged in and has at least one notification.
Inputs	Preference change; Notification action: Mark as read
Steps	1. Go to Settings . 2. Change a preference. 3. Save settings. 4. Mark a notification as read.
Expected Output	Preference is saved. Notification read status updates and unread count decreases.

TC-16: Ask AI Assistant a Question

Field	Details
Related Use Case	UC-11
Objective	Verify that the AI assistant accepts a user question and returns a response.
Preconditions	User is logged in and backend is running.
Inputs	Question: How much do I owe?
Steps	1. Go to AI Assistant . 2. Type the question. 3. Submit the message.
Expected Output	The user's question appears in the chat. The assistant returns a response or displays an error if the request cannot be completed.

4. Traceability Matrix

Test Case ID	Test Case Name	UC-01	UC-02	UC-03	UC-04	UC-05	UC-06	UC-07	UC-08	UC-09	UC-10	UC-11
TC-01	Register a New User Account	X										
TC-02	Reject Login with Invalid Credentials	X										
TC-03	Create a New Group with Invite		X			X						
TC-04	Accept a Pending Group Invite			X								
TC-05	Add an Equal Split Expense				X		X					
TC-06	Reject Invalid Percent Split Expense				X							
TC-07	Create Vote for Expense Above Threshold					X						
TC-08	Approve a Pending Expense Vote					X	X					
TC-09	Decline a Pending Expense Vote					X						
TC-10	View Dashboard Balances						X					
TC-11	Record a Settlement Payment							X				
TC-12	Create a Savings Challenge								X			
TC-13	Add Contribution to Savings Challenge								X			

Test Case ID	Test Case Name	UC-01	UC-02	UC-03	UC-04	UC-05	UC-06	UC-07	UC-08	UC-09	UC-10	UC-11
TC-14	View Analytics									X		
TC-15	Update Settings and Mark Notification Read										X	
TC-16	Ask AI Assistant a Question											X

5. Coverage Summary

Use Case ID	Use Case Name	Covered By
UC-01	Create Account and Authenticate	TC-01, TC-02
UC-02	Create and Manage Group	TC-03
UC-03	Join Existing Group	TC-04
UC-04	Log Expense with Split Options	TC-05, TC-06
UC-05	Vote on Proposed Expense	TC-03, TC-07, TC-08, TC-09
UC-06	View Dashboard and Balances	TC-05, TC-08, TC-10
UC-07	Settle Outstanding Balance	TC-11
UC-08	Create and Contribute to Challenge	TC-12, TC-13
UC-09	View Analytics and Insights	TC-14
UC-10	Update Settings and Notifications	TC-15
UC-11	Chat with AI Assistant	TC-16